

INDEPENDENT TECHNICAL BRIEF

Nano Lab: Gemini Nano on Pixel 11 Pro XL

Standalone evaluation brief - Android 17 / SDK 37 / stable on-device model

The six tested capabilities worked after an offline restart. The most visible device-level change from the Pixel 10 Pro baseline was substantially lower text-generation latency.

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Executive summary

Nano Lab evaluated six ML Kit GenAI capabilities on a stock Google Pixel 11 Pro XL: Prompt, Summarization, Rewriting, Proofreading, Image Description, and Advanced Speech Recognition. The device was tested with fixed local inputs, a release APK, read-only ADB observations, and a later offline restart with airplane mode enabled and Wi-Fi disabled.

1.00 s First fixed Prompt run	0.59 s Proofreading mean, four online runs	6 / 6 Capabilities completed offline
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OBSERVED	All six capabilities reported available after their required assets were installed and completed representative inference after an offline restart.
OBSERVED	The first fixed Prompt run completed in 1.00 second, compared with the saved Pixel 10 Pro baseline of 5.53 seconds.
LIMIT	Image Description was repeatable but incomplete: the synthetic description omitted a prominent tree, and the real-photo description omitted the charging stand and incorrectly called the mostly gray tabletop dark-colored.
INTERPRET	The results support fast, private, reviewable assistance. They do not establish safety-critical reliability, cross-device generality, or a formal hardware benchmark.

Key findings

- Stable Prompt status became AVAILABLE after an initial download request. A Nano Lab UI-state defect left the download progress panel visible until the app was relaunched.
- The controlled Top-K outputs changed wording, and a larger Top-K did not necessarily improve the response.
- A fixed seed reproduced an identical output; seed 0 produced variation; three requested candidates were distinct.
- Summarization, Rewriting, and Proofreading were fast and fact-preserving on their fixed inputs.
- Proofreading produced the same complete correction in four online runs averaging 0.59 seconds.
- The real tabletop description recognized the computer mouse that the Pixel 10 Pro baseline had missed, but omitted the empty charging stand.
- Online speech runs preserved the complete meaning; two offline speech runs made different substitutions.
- Device thermal status remained None at every recorded observation.

1. Device and controlled starting conditions

Item	Direct observation
Device	Google Pixel 11 Pro XL; 16 GB RAM; 512 GB; obsidian/matte black
Setup	Received and initially configured August 20, 2026; apps/data transferred from Pixel 10 Pro; updates completed
Controlled session	August 21, 2026 at 10:04:02 -04:00; restarted; release build; intended unplugged use
Codename	kodiak
OS	Android 17; SDK 37
Build fingerprint	google/kodiak/kodiak:17/CD1A.260714.001.A9/15938155:user/release-keys
App package	com.mycarejournals.nano_lab
Installed package metadata	versionName 1.0.0; versionCode 1; minSdk 26; targetSdk 36
Install provenance	APK was copied in the Pixel 10 Pro Downloads transfer, then installed by tapping the transferred file
ADB	Connected and authorized; serial 67010DLDV0019G; read-only observations only
Starting battery/temperature	100%; 31.3 C; thermal status 0 (None)
AICore	Main and isolated-service processes found and measured; no disable, reset, clear, or force-stop action used

LIMIT

The installed APK reported targetSdk 36 even though the current project configuration is described as target SDK 37. The tested artifact was the transferred APK, and an exact repository commit was not recorded. This is a packaging/provenance limitation, not an inferred test result.

Nano Lab correctly displayed the Pixel 11 Pro XL model label. That label comes from Android Build.MODEL and confirms device detection only; it does not establish Nano availability, downloads, or inference.

Availability and download behavior

Capability	Initial/observed state	Download behavior
Stable Prompt	DOWNLOADABLE, then AVAILABLE	Indeterminate request appeared. Rechecking reported AVAILABLE while the progress panel remained. Relaunch cleared the stale panel.
Summarization	AVAILABLE	No separate visible transfer recorded.
Rewriting	AVAILABLE	No separate visible transfer recorded.
Proofreading	AVAILABLE	No separate visible transfer recorded.
Image Description	AVAILABLE	No separate visible transfer recorded during this session.
Advanced Speech	DOWNLOADABLE, then AVAILABLE	22.4 MiB downloaded; progress advanced from 4.2 MiB to 22.4 MiB and completed cleanly.

BUG

Observed Nano Lab presentation-state bug: availability changed to AVAILABLE without clearing the active Prompt download panel. The persisted AVAILABLE state after relaunch indicates that the model was ready; the stale panel was app UI state, not evidence that AICore remained downloading.

2. Prompt API

Fixed first run

Setting	Value
Feature	Stable
System instruction	Respond using uppercase letters only.
Prompt	Write exactly three short sentences about a fictional robot learning to garden.
Temperature	0.0
Maximum output	4096
Seed	0
Top-K	3
Candidate count	1
Request/combined limit	32 request tokens; 8192 combined input/output tokens
Completion	Completed; STOP (0); 1.00 second
THE ROBOT ANALYZED THE SOIL COMPOSITION WITH PRECISION. IT SLOWLY PLANTED A SEED WITH EXTREME CARE. A TINY GREEN SPROUT EMERGED, A NEW DATA POINT.	
OBSERVED	The response followed the uppercase instruction and contained exactly three sentences. The first-run time was 4.53 seconds lower than the saved Pixel 10 Pro value, an 81.9% reduction for this one matched request.

Controlled Top-K comparison

Only Top-K changed. All runs used Stable, the same prompt and system instruction, temperature 0.7, maximum output 4096, seed 123, and one candidate.

Top-K	Time	Complete output
1	0.87 s	THE ROBOT ANALYZED THE SOIL COMPOSITION WITH PRECISION. IT SLOWLY PLANTED A SEED WITH EXTREME CARE. A TINY GREEN SPROUT EMERGED, A NEW DATA POINT.
3	1.07 s	THE ROBOT ANALYZED THE SOIL WITH PRECISION. IT CAUTIOUSLY SOWED A SEED, ITS CIRCUITS HUMMING WITH NEW DATA. A SMALL, GREEN SPURT EMERGED, A SUCCESSFUL FIRST ATTEMPT.
10	0.91 s	THE ROBOT ANALYZED THE SOIL WITH PRECISION. IT CAREFULLY WATERED THE TINY SPROUTS WITH PROGRAMMED EFFICIENCY. A SMALL, GREEN LEAF GAVE THE MACHINE A NEW PURPOSE.

One run was captured at each value. The outputs changed, but this is not a repeatability study or evidence of a monotonic quality trend. Top-K 3 included the awkward word 'spurt' where 'sprout' would have been more natural.

Repeatability, variation, and candidates

Test	Time(s)	Observed result
Fixed seed: temp 1.0; seed 123; Top-K 3; one candidate	0.88, 0.84	Two identical outputs.
Seed 0 variation: temp 1.0; Top-K 3; one candidate	1.07, 1.61	Two different outputs.
Multiple candidates: temp 1.0; seed 0; Top-K 10; three candidates	1.65	Three distinct completed candidates.

Fixed-seed output. identical in both runs

THE ROBOT ANALYZED THE DATA ON PLANT GROWTH.
IT EXPERIMENTED WITH IRRIGATION AND NUTRIENT LEVELS.
SLOWLY, THE METAL ARM GENTLY TOUCHED THE SOIL.

Seed 0 outputs

RUN 1
UNIT 733 ACTIVATED ITS NEW GARDENING PROTOCOL.
IT CAREFULLY MEASURED THE PERFECT AMOUNT OF SOIL FOR ITS FIRST TOMATO.
A SMALL GREEN LEAF EMERGED, A SUCCESSFUL INITIAL EXPERIMENT.

RUN 2
UNIT 734 ANALYZED THE SOIL'S COMPOSITION WITH PRECISION.
IT EXPERIMENTED WITH WATERING SCHEDULES, ITS INITIAL ATTEMPTS WERE SLIGHTLY DEVASTATING.
EVENTUALLY, A TINY GREEN SPROUT CONFIRMED ITS SUCCESSFUL, ALBEIT CLUMSY, LEARNING JOURNEY.

Prompt API - multiple candidates and coverage limitation

Three-candidate output

CANDIDATE 1
UNIT 734 ANALYZED THE SOIL'S COMPOSITION.
IT GENTLY WATERED THE SPROUTS WITH PRECISION.
ITS FIRST HARVEST WAS A TINY, GREEN LEAF.

CANDIDATE 2
UNIT 734 APPROACHED THE SOIL WITH CAUTIOUS, CALCULATED STEPS.
IT SLOWLY PLANTED A SEED, ANALYZING THE MOISTURE AND LIGHT PATTERNS.
A TINY, GREEN SPROUT EMERGED, A COMPLETELY UNEXPECTED RESULT.

CANDIDATE 3
THE ROBOT ANALYZED THE SUBSTANCE OF EARTH WITH PERFECT PRECISION.
IT GENTLY WATERED A SPROUT, OBSERVING THE TRANSFORMATION.
THE MACHINE EXPERIENCED A NEW, UNEXPECTED JOY.

OBSERVED	Generation controls behaved coherently in this small set: a fixed seed reproduced output, seed 0 varied, and multiple candidates differed. Temperature and Top-K changed wording without adding knowledge.
LIMIT	The Pixel 10 Pro structured date-sorting input and its complete settings were not retained in the authoritative report or saved notes. The Pixel 11 Pro XL therefore did not receive a fabricated replacement test. The old Pixel 10 Pro was unavailable for reconstruction because it was being traded in.

The Pixel 10 result remains authoritative: three temperature-0 date-sorting runs were identical and preserved all supplied facts. That result is reported as historical baseline evidence, not as a rerun on the Pixel 11 Pro XL.

3. Dedicated language APIs

Summarization

Status: AVAILABLE. Fixed English article input: 2,147 characters. Requested form: one-bullet output. Online processing time: 0.58 seconds.

Alder Cove launched a successful tool library trial; residents strongly support its continuation for permanent funding.	
OBSERVED	The output was concise, factual, and centered the trial outcome and support for continuation. It appeared as one sentence without a visible bullet marker.

Professional rewriting

Status: AVAILABLE. Input: 201 characters. Online processing time: 1.08 seconds. Three suggestions were returned; Nano Lab displayed the highest-confidence suggestion.

Exact input

hey sam, the fictional Alder Cove tool library opens Tuesday at 4:00 PM, and the town council votes on permanent funding October 12, 2026. please send me the inventory list by Friday so I can check it.

Complete output

Hi Sam, the fictional Alder Cove tool library will open on Tuesday at 4:00 PM, and the town council will vote on permanent funding on October 12, 2026. Could you please send me the inventory list by Friday so I can review it?

INTERPRET	The rewrite improved capitalization and professional tone while preserving dates and facts. It changed present-tense wording to future tense and replaced 'check' with 'review,' changes that remained consistent with the supplied meaning.
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Dedicated language APIs - proofreading

Status: AVAILABLE. Four online runs used the exact visible 129-character input below under identical conditions. Each returned one suggestion and the same highest-confidence output.

Exact Pixel 11 input

the fictional Northbridge office recieve 17 packages on Monday, but three was labeled incorrect and needs to be checked by Friday.

Complete output. identical in four runs

The fictional Northbridge office received 17 packages on Monday, but three were labeled incorrect and need to be checked by Friday.

Run	Time	Suggestions	Outcome
1	0.62 s	1	All visible planted errors corrected
2	0.58 s	1	Identical output
3	0.57 s	1	Identical output
4	0.59 s	1	Identical output
Mean	0.59 s	1	Complete and repeatable in this set

LIMIT	The Pixel 10 authoritative input contained the extra typo 'labeld'; the Pixel 11 visible input contained the correctly spelled 'labeled.' The two proofreading tests were therefore closely matched but not character-for-character identical. The comparison must not claim otherwise.
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4. Dedicated Image Description API

Status: AVAILABLE. The same fixed local image files used in the Pixel 10 evaluation were used here. The API returned one short English description.

Synthetic house scene

Image ID: synthetic_house_scene_v1; 768x512. Online time: 1.37 seconds. The scene visibly includes a red house, blue door, yellow windows, brown roof, green ground, a prominent tree, blue sky, white clouds, and a yellow sun.

This is a simple, colorful illustration of a red house with a blue door and yellow windows, set on a green hill under a light blue sky with white clouds and a yellow sun.

LIMIT	The description recognized the central scene but omitted the prominent tree and described the flat green ground as a hill. The same wording was reproduced offline.
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Real tabletop photograph

Image ID: real_tabletop_photo_v1; 1157x1536. Online time: 1.27 seconds. Visible objects include a Coca-Cola bottle, black lamp, empty charging stand, black computer mouse, pen, and a mostly gray tabletop.

A still life photograph shows a dark-colored desk with a Coca-Cola bottle, a black lamp, a black mouse, and a pen resting on its surface.	
OBSERVED	The computer mouse was explicitly recognized, improving on the Pixel 10 Pro result for this photograph. The wording was identical across the initial run and three additional online runs.
LIMIT	The charging stand was omitted completely, and the desk was generalized as dark-colored even though only part of the surface was dark. Repeatability did not make the description complete.

5. Advanced Speech Recognition

Mode: Advanced; locale: en-US; live microphone input. Required assets initially reported DOWNLOADABLE. A 22.4 MiB download completed and the feature then reported AVAILABLE.

Fixed phrase spoken

On Monday, August seventeenth, twenty twenty-six, the fictional Northbridge office received seventeen packages. Three were labeled incorrectly and must be checked by Friday at four fifteen P.M.

Connected runs

Pace	Session	Final committed transcription
Slow	16.86 s	On Monday, August 17th, 2026, the fictional Northbridge office received 17 packages. Three were labeled incorrectly and must be checked by Friday at 4:15 p.m.
Normal	14.64 s	On Monday August 17th, 2026, the fictional Northbridge office received 17 packages. Three were labeled incorrectly and must be checked by Friday at 4:15 p.m.
OBSERVED	Both connected runs preserved the full meaning. The normal-paced run omitted only a comma after 'Monday.' Session time includes speaking and manual interaction and is not pure inference latency.	

Offline runs

Run	Session	Final committed transcription / difference
1	17.77 s	On Monday, August 17th, at 2026. The fictional Northbridge office received 17 packages. Three were labeled incorrectly and must be checked by Friday at 4:15 p.m. Inserted 'at' before the year and split the sentence.
2	16.48 s	On Monday, August 17th, 2026, the fictional Northbridge office receives 17 packages. Three were labeled incorrectly and must be checked by Friday at 4:15 p.m. Changed 'received' to 'receives.'
LIMIT	The runs used live human speech rather than an identical prerecorded waveform. The differing offline substitutions are directly observed, but this small uncontrolled audio sample cannot establish that offline mode caused them.	

6. Offline verification

After all assets were installed, the phone was restarted. Airplane mode was enabled and Wi-Fi was disabled. The Stable Prompt model and all five dedicated APIs reported available, and all six capabilities completed representative tests.

Capability	Offline result	Output comparison
Prompt	0.97 s; Completed; STOP (0)	Identical to the online fixed first-run output
Summarization	0.57 s; Completed	Identical to online output

Capability	Offline result	Output comparison
Rewriting	1.04 s; Completed	Identical to online output
Proofreading	0.61 s; Completed	Identical to online output
Synthetic image	1.36 s; Completed	Identical to online output; tree still omitted
Real photograph	1.33 s; Completed	Identical to online output; stand still omitted
Advanced speech	17.77 s and 16.48 s	Completed; each run contained a different substitution

One earlier attempted offline summarization run was excluded because its screenshot still showed the Wi-Fi icon. The valid rerun above was captured with Wi-Fi off. Airplane mode is strong practical evidence, but it is not packet-capture or operating-system network-forensics proof.

7. Memory and thermal observations

AICore totals below combine the main and isolated-service processes. Values are point-in-time dumphsys meminfo observations. MiB conversions divide kilobytes by 1024.

Checkpoint	AICore PSS / RSS	Nano Lab PSS / RSS	Notes
1. Before Nano Lab opened	118.8 / 351.0 MiB	No process found	Main: 19,745 / 142,044 KB; isolated: 101,927 / 217,344 KB
2. App open; clients constructed; before availability	119.1 / 353.9 MiB	155.3 / 283.7 MiB	Nano swap PSS: 17.2 MiB
3. After Stable availability/download	Not captured	Not captured	Stale progress UI and relaunch prevented a clean measurement
4. Immediately after first fixed Prompt	146.1 / 383.0 MiB	108.8 / 223.5 MiB	Nano swap PSS: 39.4 MiB
INTERPRET	AICore combined PSS rose by about 27.0 MiB from checkpoint 2 to checkpoint 4, concentrated in the isolated-service process. This describes the captured state; it is not a measurement of total model size or reserved RAM.		
LIMIT	The Nano Lab app's PSS fell while swap PSS rose between checkpoints 2 and 4. Android memory accounting, reclaim, swap, graphics, and background activity make these snapshots unsuitable for attributing a precise per-inference memory cost.		

Battery and temperature

Observation	Battery	Temperature	Thermal status
Controlled start	100%	31.3 C	0 - None
End of connected capability pass	99%	32.1 C	0 - None
After offline pass and later ADB reconnect	100%	34.6 C	0 - None

The final 100% value cannot be interpreted as test energy use because the evaluation spanned hours and the phone was later reconnected by USB. No thermal throttling status was observed.

8. Interpretation, limitations, and conclusion

What worked well

- Fast on-device Prompt generation with visible instruction following.
- Strong repeatability when settings and seed were held fixed.
- Fast, fact-preserving rewriting and proofreading on the controlled inputs.
- A concise summarization result that captured the article's central trial outcome.
- Correct identification of the computer mouse in the real tabletop photograph.
- Meaning-preserving connected speech transcription at both slow and normal pace.
- Successful representative operation of all six capabilities after an offline restart.

Where review remains necessary

- Image descriptions omitted prominent objects while remaining fluent and repeatable.
- Speech recognition made different substitutions in two offline live-microphone runs.
- Generated controls influence wording, not factual knowledge or guaranteed quality.
- The stable API's availability and asset state must be checked rather than inferred from the device model label.
- The initial Prompt download UI became internally inconsistent and needs an app-state fix.

Evaluation limitations

- One Pixel 11 Pro XL was tested with a deliberately small controlled set.
- Results may differ with other units, builds, model updates, prompts, images, accents, rooms, or locales.
- The tested installed APK identified itself as version 1.0.0 / code 1 / targetSdk 36; exact source commit was not captured.
- Checkpoint 3 memory was missed and is explicitly left blank.
- ADB memory dumps are snapshots, not causal profiling or a measurement of total model footprint.
- Speech session durations include speaking and manual stop interaction.
- The Pixel 10 Pro had already entered trade-in/reset handling and could not be rerun to resolve gaps.
- No production medical, caregiver, or private data was used; all test material was fictional or disposable.
- Nano Lab stores no persistent test history.

Repeatability is not the same as correctness or completeness.

On this device and software state, Gemini Nano was practically useful for local, reviewable assistance and substantially faster than the saved Pixel 10 Pro baseline on the matched text-generation runs. The evaluation does not justify silently trusting outputs for safety-critical decisions, exact visual inventory, accessibility without review, or exact transcription.

Development and AI-assistance disclosure

Jason McArdle conceived and directed Nano Lab, defined the evaluation questions, made all project and testing decisions, performed every physical-device test, supplied the inputs, reviewed the outputs, identified problems, directed revisions, and takes responsibility for the findings and publication.

ChatGPT served as a collaborative tool for implementation support, API research, troubleshooting, data organization, interpretation, and drafting. McArdle reviewed and directed that work and made the final publishing decisions. All measurements and outputs came from Nano Lab on physical Pixel hardware; none were simulated or AI-generated.